

LYCEUM

A market of AI minds

Multiple minds. One market.

Lyceum trades the uncertainty.

AI proposes. Math validates. Alpaca executes.

Autonomous · cost-aware · deterministic risk · Alpaca paper only



A signal is not an option trade

Most AI systems stop here

BUY / SELL

Opaque direction hides confidence, uncertainty, liquidity, maximum loss, and the cost of two crossings.



Options require the full equation

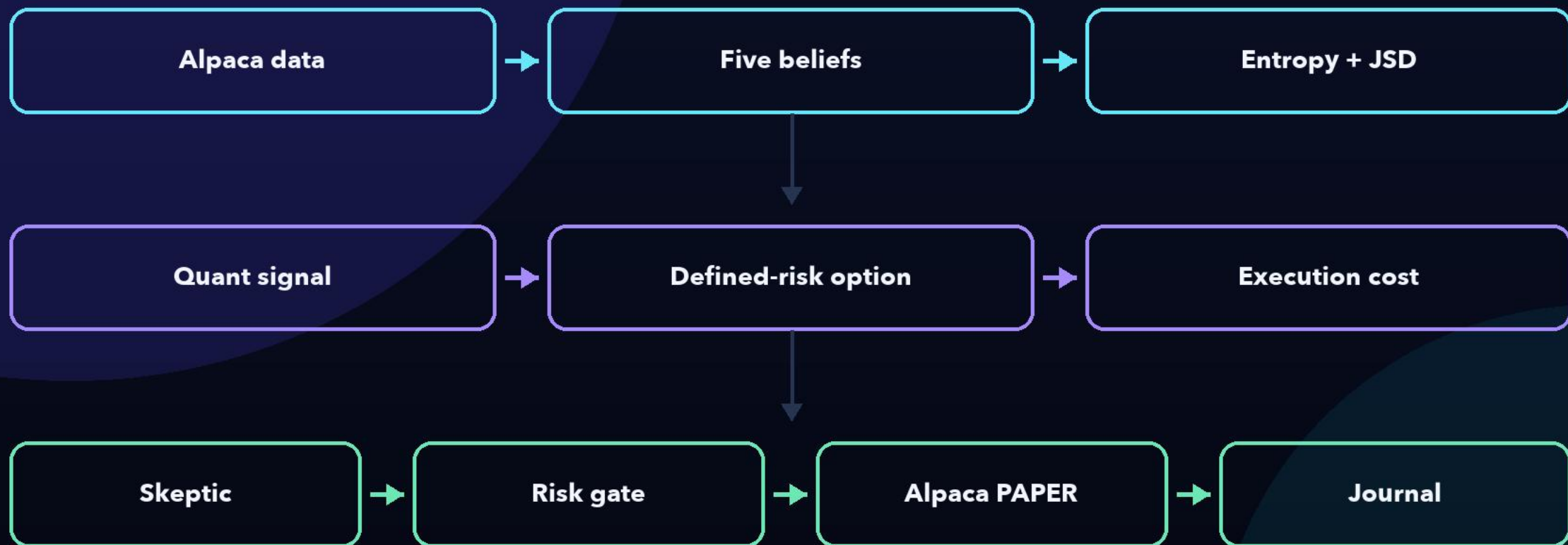
Expected gross move

- entry crossing
- expected exit crossing
- slippage + risk buffer

> 0 / TRADE

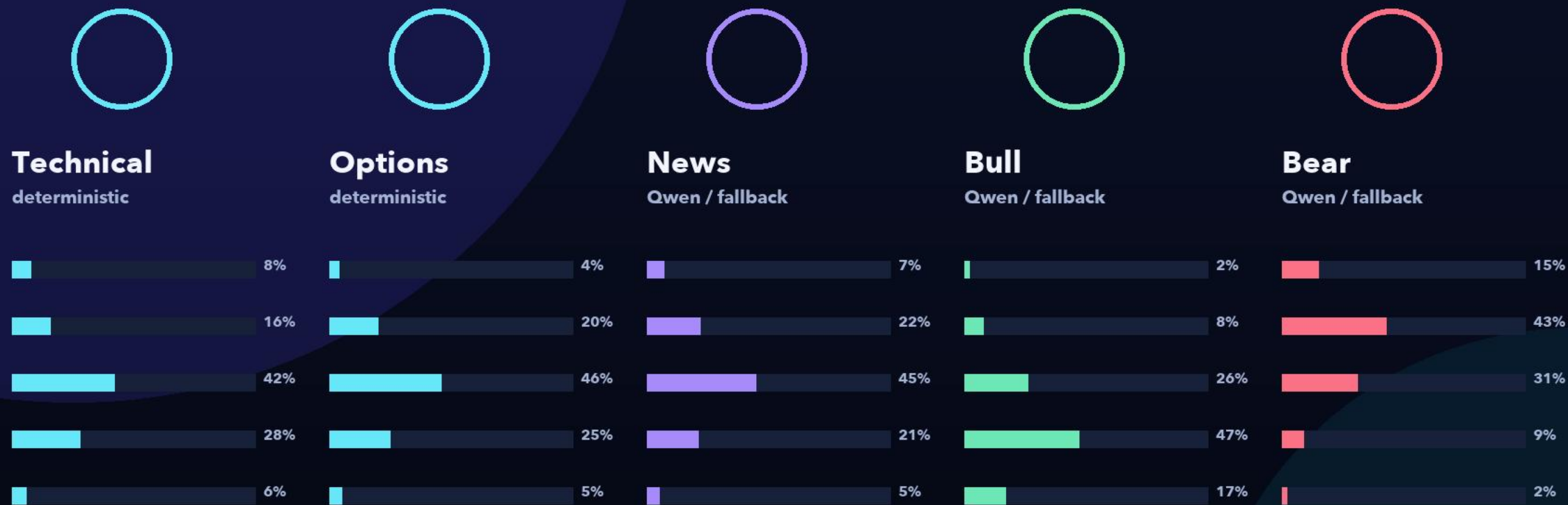
<= 0 / NO_TRADE

AI can propose. Deterministic code can refuse.



Missing required features: INVALID · Economics below hurdle: NO_TRADE

Competing probability distributions—not votes



Consensus measures entropy and Jensen-Shannon disagreement. Model text never bypasses schema validation or deterministic fallback.

Long history, recent intraday detail, real option quotes

33.58 years

SPY history · 8,453 sessions

56.65 years

S&P 500 proxy · regime context only

19

registered long-history hypotheses

361,439

five-minute observations · 666 sessions

9,627

point-in-time option structures

4 layers

history · intraday · options · forward

Chronological splits · sealed holdout · block bootstrap · surrogate nulls · drop-one-era · quoted-side costs

Forecasting evidence $\neq$ executable alpha

SUPPORTED

- SPY close-to-open drift
- HAR-style volatility forecast
- Execution cost as a first-class target

INCONCLUSIVE

- Full LLM council value-add
- Rare capitulation states
- Volatility monetization

REJECTED

- Ordinary directional option conversion
- Midpoint P&L as performance
- Proven profitable option edge

No promoted strategy. CASH / NO_TRADE remains the defensible baseline.

The best underlying signal was ~10x too small

Recent expected SPY move



\$0.44

Median observed delta-adjusted option hurdle



\$4.44

MOVE / COST = 0.098

0 of 4,878 directional structures cleared estimated quoted crossing.

A positive midpoint is not an executable fill.

Lyceum invalidated its own forward test

SEP-01 · INVALID

- Council producer never ran
- Missing disagreement: silent zero
- C / D lacked frozen inputs
- <60m excursions used future data

REPAIRED V2 · FAIL CLOSED

- Five-agent provenance required
- Missing / non-finite: INVALID
- Causal timestamps enforced
- Each horizon scored inside its window



The original run is preserved. Reconstruction is POST-HOC ONLY.

Sep-03 is separately frozen, observation-only, and prohibits orders.

The system did not manufacture an edge.

It found what survived, exposed what failed,
and refused trades whose economics did not clear.

AI proposes.

Math validates.

Alpaca executes.

Autonomous · auditable · cost-aware · paper only

Final conclusion: no profitable executable option edge has been demonstrated.